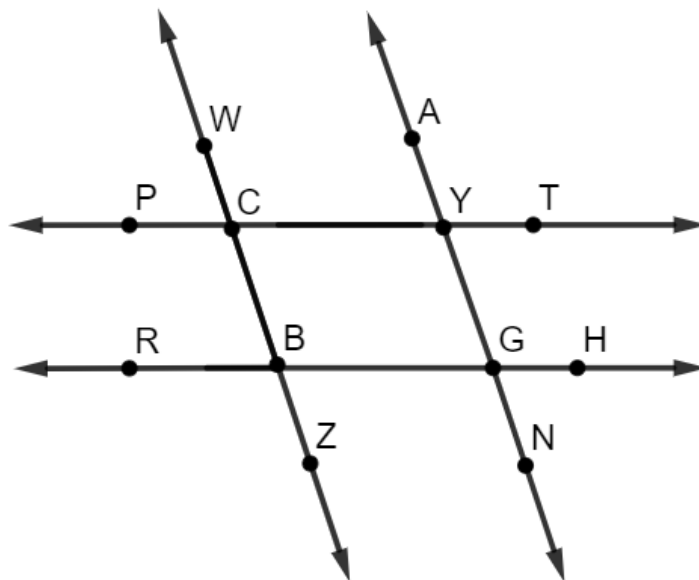


3.9.2 Activity

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$,
 $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$

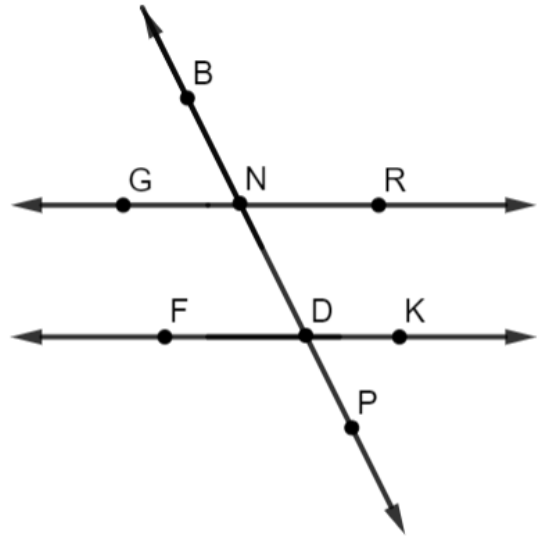


Statement	Reason
$\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	Given
$\angle WCY \cong \angle YGH$	Given
$\angle WCY \cong \angle AYT$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\angle AYT \cong \angle YGH$	Transitive Property of Congruence
$\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	If two lines are intersected by a transversal so that the corresponding angles are congruent, then the lines are parallel.

Proof B

Given: $\angle GND$ and $\angle PDK$ are supplementary

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

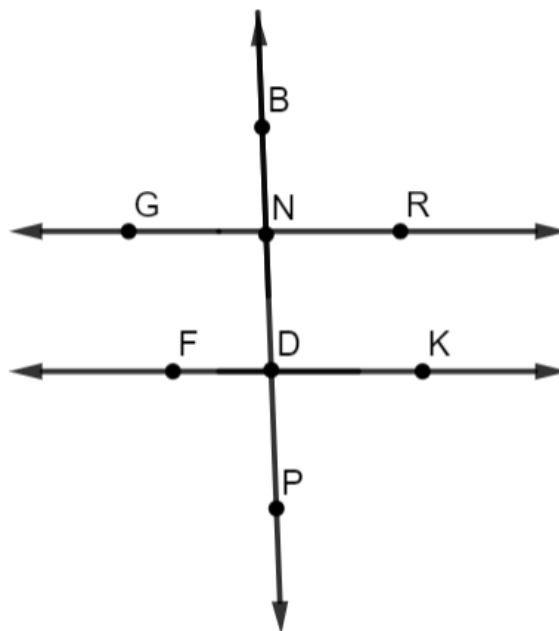


Statement	Reason
$\angle GND$ and $\angle PDK$ are supplementary	Given
$m\angle GND + m\angle PDK = 180^\circ$	Definition of Supplementary Angles
$\angle GND \cong \angle BNR$	Vertical Angles are Congruent
$m\angle GND = m\angle BNR$	Definition of Congruence
$m\angle BNR + m\angle PDK = 180^\circ$	Substitution Property of Equality
$\angle BNR$ and $\angle PDK$ are supplementary	Definition of Supplementary Angles
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	If two lines are intersected by a transversal so that the consecutive exterior angles are supplementary, then the lines are parallel.

Proof C

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

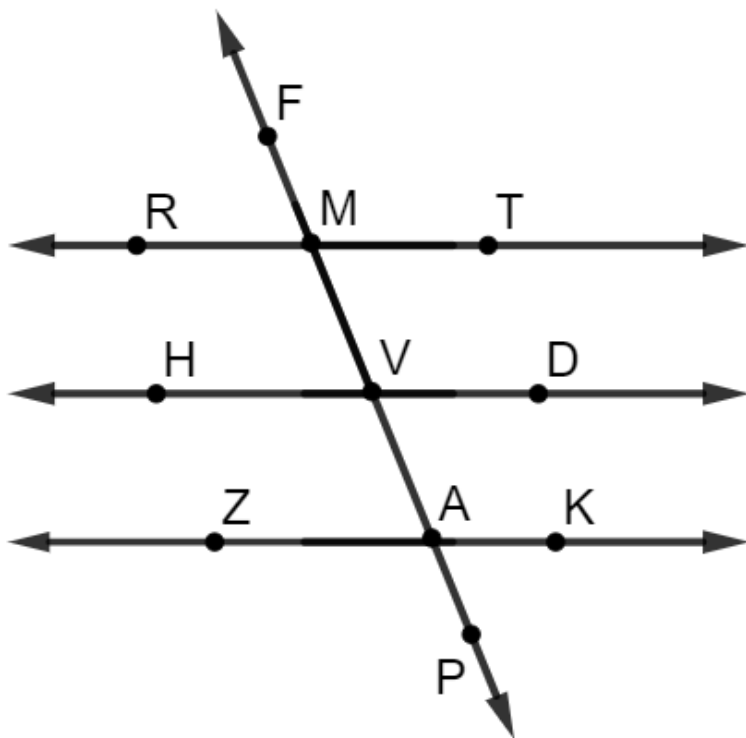


Statement	Reason
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	Given
$\angle GNB$ is a right angle	Definition of Perpendicular Lines
$m\angle GNB = 90^\circ$	Definition of Right Angle
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	Given
$\angle GNB \cong \angle FDN$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$m\angle GNB = m\angle FDN$	Definition of Congruence
$m\angle FDN = 90^\circ$	Substitution Property of Equality
$\angle FDN$ is a right angle	Definition of Right Angle
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	Definition of Perpendicular Lines

Proof D

Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$,
 $\angle TMV \cong \angle AVD$

Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$

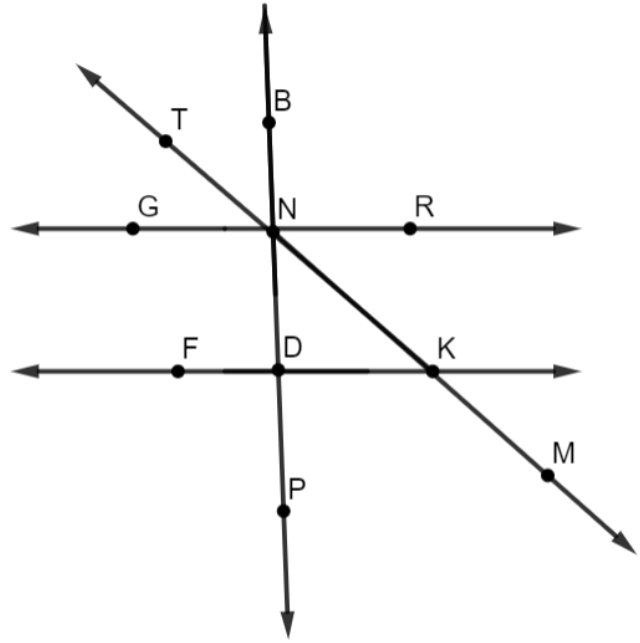


Statement	Reason
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$	Given
$\angle TMV \cong \angle KAP$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\angle TMV \cong \angle AVD$	Given
$\angle KAP \cong \angle AVD$	Transitive Property of Congruence
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$	If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.

Proof E

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

Prove: $\angle DKM$ and $\angle GNT$ are supplementary

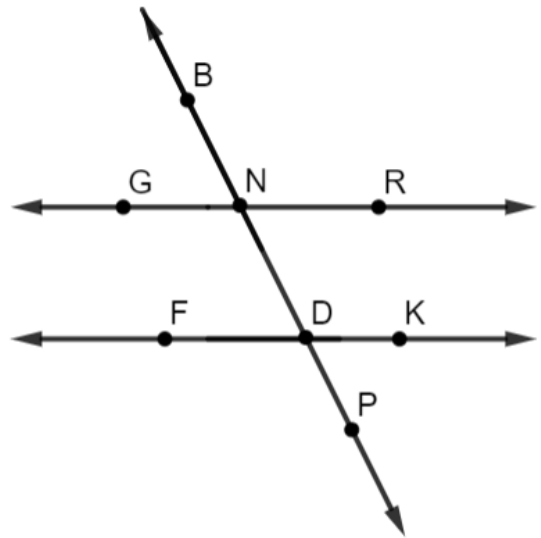


Statement	Reason
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	Given
$\angle BNR$ is a right angle	Definition of Perpendicular Lines
$m\angle BNR = 90^\circ$	Definition of Right Angle
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	Given
$\angle NDK$ is a right angle	Definition of Perpendicular Lines
$m\angle NDK = 90^\circ$	Definition of Right Angle
$m\angle BNR = m\angle NDK$	Substitution Property of Equality
$\angle BNR \cong \angle NDK$	Definition of Congruence
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.
$\angle DKM$ and $\angle GNT$ are supplementary	If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.

Proof F

Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\angle GND$ and $\angle PDK$ are supplementary

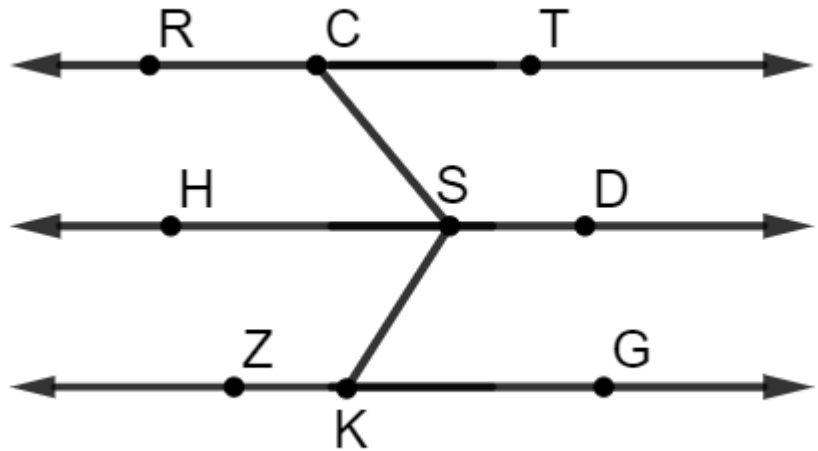


Statement	Reason
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	Given
$\angle GND$ and $\angle FDN$ are supplementary	If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.
$m\angle GND + m\angle FDN = 180^\circ$	Definition of Supplementary
$\angle FDN \cong \angle PDK$	Vertical Angles are Congruent
$m\angle FDN = m\angle PDK$	Definition of Congruence
$m\angle GND + m\angle PDK = 180^\circ$	Substitution Property of Equality
$\angle GND$ and $\angle PDK$ are supplementary	Definition of Supplementary Angles

Proof G

Given: $\angle TCS \cong \angle HSC$,
 $\angle HSK \cong \angle SKG$

Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$

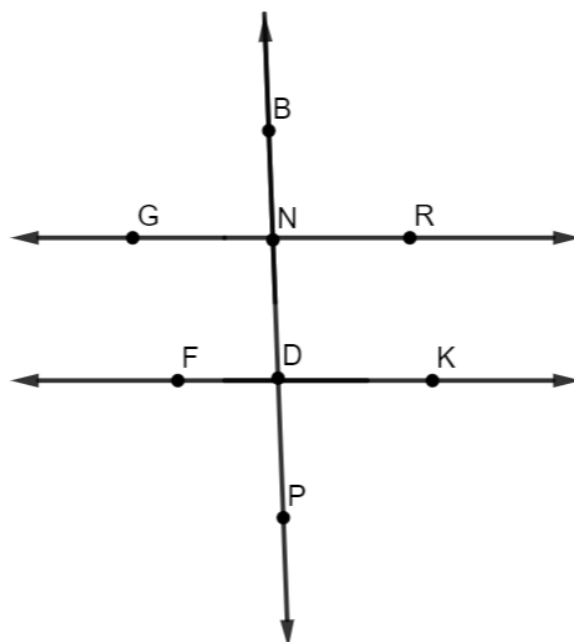


Statement	Reason
$\angle TCS \cong \angle HSC$	Given
$\overleftrightarrow{RT} \parallel \overleftrightarrow{HD}$	If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.
$\angle HSK \cong \angle SKG$	Given
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZG}$	If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$	Transitive Property of Parallel Lines

Proof H

Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

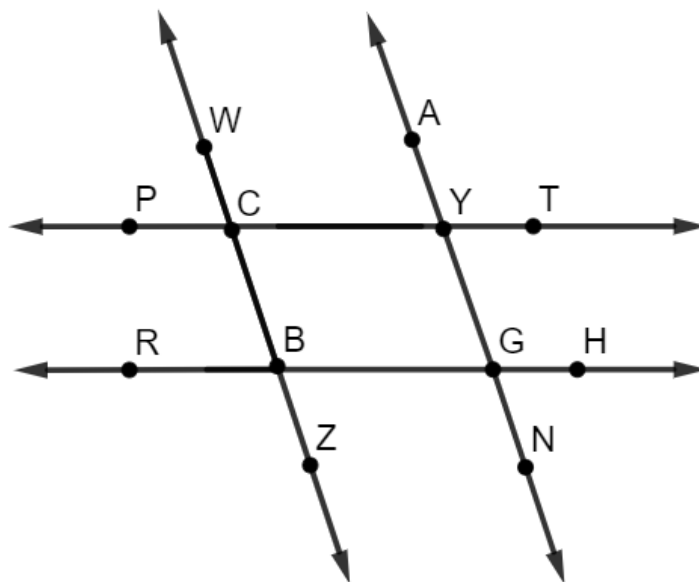


Statement	Reason
$\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$	Given
$\angle BNR$ is a right angle. $\angle NDK$ is a right angle.	Definition of Perpendicular Lines
$m\angle BNR = 90^\circ$ $m\angle NDK = 90^\circ$	Definition of a right angle
$m\angle BNR = m\angle NDK$	Transitive Property of Equality
$\angle BNR \cong \angle NDK$	Definition of Congruence
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$,
 $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$

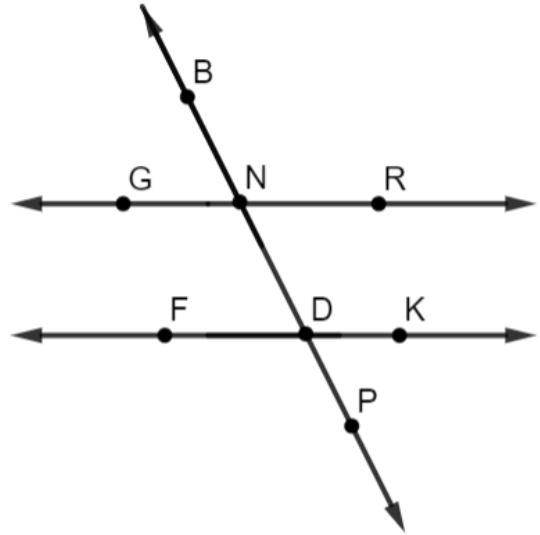


Statement	Reason

Proof B

Given: $\angle GND$ and $\angle PDK$ are supplementary

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

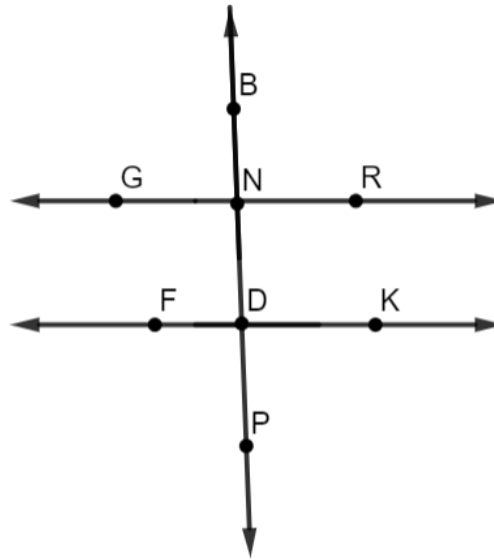


Statement	Reason

Proof C

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

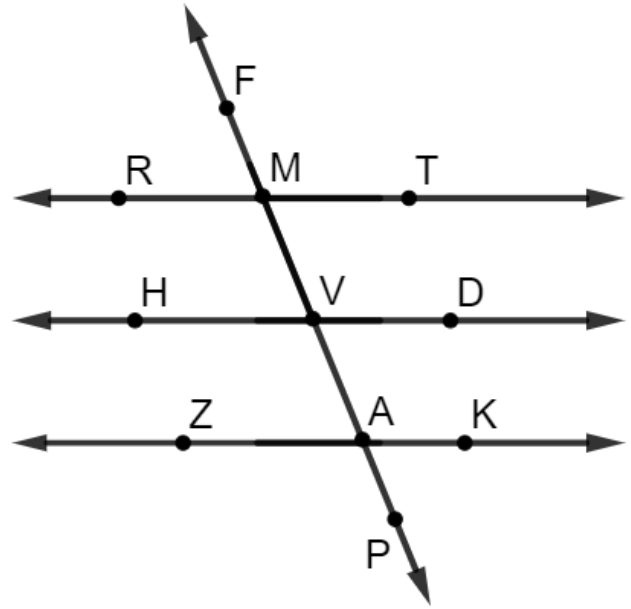
Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

[illegible]

Proof D

Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$,
 $\angle TMV \cong \angle AVD$

Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$

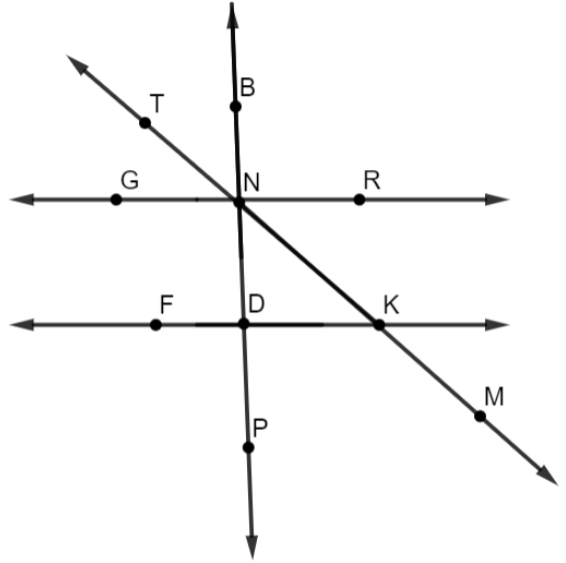


Statement	Reason

Proof E

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

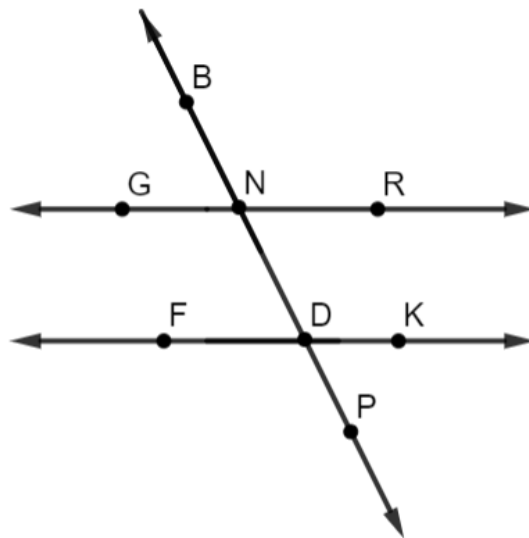
Prove: $\angle DKM$ and $\angle GNT$ are supplementary

[illegible]

Proof F

Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

Prove: $\angle GND$ and $\angle PDK$ are supplementary

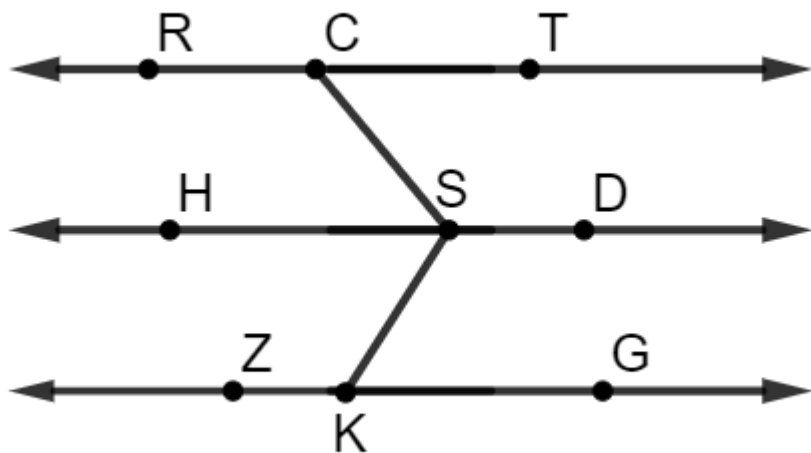


Statement	Reason

Proof G

Given: $\angle TCS \cong \angle HSC$,
 $\angle HSK \cong \angle SKG$

Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$

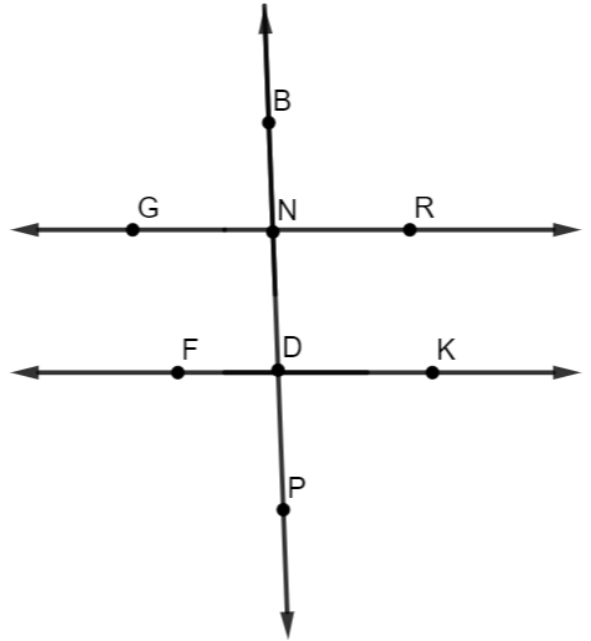


Statement	Reason

Proof H

Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



Statement	Reason